



Module	Portfolio Assessment	Type
Collaborative Development (5CS024)	1	Individual Report

Academic Management System (AMS)

INDIVIDUAL REPORT: SPRINT 1

Student Name: Manavi Rai

Student ID:2510725

Section : L5CG11

Module: Collaborative Development (5CS024)

Project: Academic Management System (AMS)

Role: Frontend Developer / UI Designer

Module Leader: Yogesh Bikram Shah

Tutor:Mr.Bibek Thapa

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Acknowledgement

"First, I am pleased to express my heartfelt thanks to my tutor and lecturer for their guidance throughout the initial stage of this assignment. Their teachings regarding agile approaches and professional programming techniques were indeed very beneficial to bridge the gap between theoretical understanding and practical application.

Second, I would like to thank my fellow team members in THE U-TEAM for encouraging and motivating me to perform to the best of my ability. Being a member of a group allowed me to know how to deal with real-world software development along with version control and task management inside a software development project."

1) Assignment Report (Project Overview & Technology Choice)

The AMS is a web application that is designed to serve as a central portal for communicating and managing data between the students, teachers, and administrative staff. As the Frontend Developer, I focused on implementing a premium Glassmorphism UI using PHP, MySQL, and CSS3.

Technological Selection Justification: In the current case study, we analyzed several technology stacks. Even though frameworks such as React might be useful, we decided to go with a combination of Vanilla CSS3 and JavaScript on the client-side to keep the source code light and have complete control over design tokens. In terms of server-side technology, PHP 8.0 was selected to implement the frontend functionality and rendering. MariaDB was selected as our DBMS, considering the need for a relational database.

Risks and limitations:

Security: PHP is at risk from SQL Injection attacks; this was avoided by using PDO prepared statements.

Complexity: The development process of AMS-76 requires more time because of its custom implementation of responsiveness.

Compatibility: Legacy browsers will not support Glassmorphism blurring, and so we have implemented solid color fallbacks.

2) Self-Appraisal Form

Student Number	2510725	Name	Manavi Rai
Project	Project Academic Management System (AMS)	Date	20/04/2026
Role	Frontend Developer	Team	THE U-TEAM
Sprint	1		

Personal objectives – performance measurement

Objectives	Evidence provided	Evaluation Student / Tutor	
Language Selection	I selected HTML5, CSS3, and JavaScript for the frontend and PHP for UI logic	/10	
Tutor feedback:			
Functional Requirements	I delivered the Signup system, Teacher Dashboard, and Responsive layouts (AMS-60 to AMS-81).	/15	
Tutor feedback:			
Version Control	I managed all my UI code and CSS assets through GitHub.	/10	
Tutor feedback:			

Deadlines	I utilized Jira to track my 7 assigned tasks and completed all of them on time.	/5	
Tutor feedback:			

Retrospective for Sprint 1

What Went Well: The UI design using Glassmorphism was consistently implemented throughout all the modules.

Challenges: Managing responsive grid layouts (AMS-76) manually was more time-consuming than expected.

Future Plans in Sprint 2: Start integrating frontend and API calls as early as possible.

3) Collaborative Form (Collaboration Document)

During the entire course of Sprint 1, I kept an extremely coordinated pace of work with my teammates. We worked together within a professional Agile environment, using Jira to manage our tasks , git for Version control and Slack/WhatsApp to communicate and solve problems instantly.

Proof of Effective Cooperation

Mainly, the objective of mine was to make sure that the user interface design is consistent among all modules of the system. For that purpose, I attended each and every sprint planning meeting as well as stand-up. The reason behind attending all these meetings was to ensure that there is consistency between the UI design and the system design. In collaboration with our team leader, Unibha Neupane, I tried to design the user interface based on the requirement of the project. With continuous coordination with backend developers, there was no room left for silo development.

Tracking Issues and Quality Assurance

Our approach was to ensure traceability of everything done within the development process. I used the Jira tool to control my tasks by assigning each commit to a unique issue ID (AMS-60 to AMS-81). With this approach, we could keep track of how healthy our project was at all times. In one particular instance, when there were issues with the layout on the high-definition mobile screens, I created an entry related to the problem in my Jira task named AMS-76 (Make dashboards responsive). From there, I made notes on how to solve the problem before moving the ticket through the 'In Progress' and 'Review' states before finalizing it.

Strategic Time Management and Fulfilling Deadlines

In order to ensure the sustainability of the project, a proactive approach was taken in the execution of my tasks. Since the tasks at hand involved high impact actions, such as signup UI (AMS-60) and authentication flow, it enabled me to finish off with my task well before the sprint review was due. My tasks (until AMS-81) were strategically completed in advance in order to give the other members ample time to perform a cross module testing. The additional time proved critical as it helped in detecting the edge cases of the user registration process.

Personal Reflection

"Looking back, I can see that being a Frontend Developer is not only about writing CSS but also about participating in the bigger Agile process. Being able to work on GitLab gave me a chance to participate in an organized process together with my team members. In the beginning, it was difficult for me to cope with all the nuances of custom responsiveness (AMS-76). Nevertheless, the cooperation and problem-solving using Jira allowed me to get a better understanding of how important it is to think about the user before considering my own likes and dislikes. I am happy with the level of professionalism that this sprint taught me."

4) Evidence of Individual Work (Technical Appendix)

a) Signup Module (AMS-60 & AMS-61)

i was solely in charge of the user registration user interface.

UI Design (AMS-60): I designed a multi-section form that made use of a “card” layout. I created my own CSS styling that grouped personal details, guardianship details, and role selection together.

Frontend Validation (AMS-61): I created a live validation system to verify the data inputted by users. I used PHP code to create a strict Regex that ensured the password had an uppercase letter, number, and symbol.

SCREENSHOT 1: Signup Page UI

AMS

Create Your Account

Register for the Academic Management System

PERSONAL DETAILS

First Name

Last Name

Date of Birth

Day

▼

Month

▼

Year

▼

Address

Email Address

Contact Number

Password

Create a strong password

Confirm Password

Confirm your password

GUARDIAN INFORMATION

Mother's Name

Father's Name

Guardian Contact Number

SELECT ROLE

Student

Teacher

Register Account

Already have an account? [Log in here](#)

CODE SCREENSHOT 1: Validation logic in signup.php

```
24 $strongPasswordRegex = "/^(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9])(?=.*[@$!%*?&]).{6,}$/";
25 $contactRegex = "/^\+?\d{10,15}$/";
26 if (!preg_match($strongPasswordRegex, $password)) {
27     $signup_error = "Password must be at least 6 characters long, include an uppercase letter, a lowercase letter, a
28 } elseif (!preg_match($contactRegex, $contact)) {
29     $signup_error = "Validation Error: Please enter a valid contact number (10 to 15 digits).";
30 } elseif (!preg_match($contactRegex, $guardianContact)) {
31     $signup_error = "Validation Error: Please enter a valid guardian contact number (10 to 15 digits).";
32 } else {
33     require_once 'includes/db.php';
```

b) Password Management UI (AMS-67 & AMS-68)

I created the user interface design for the sections of the app related to security features.

Password Recovery (AMS-67): This is the “Forgot Password” feature, which allows users to input their emails to get the OTP.

Enter New Password (AMS-68): This is where users would enter their new passwords post-verification, meeting all the criteria regarding security.

SCREENSHOT 2: Password Reset/Recovery UI

AMS
Recover Your Account
We'll send a secure reset link to your registered email address.

Email Address
raimanavi66@gmail.com

Send Reset Link

Remembered your password? [Back to Login](#)

2.1:

AMS

Create New Password

Ensure your account is secure with a strong password.

New Password

Min. 6 characters

Confirm Password

Repeat new password

[Back to Login](#)

Update My Password

INSERT CODE SCREENSHOT 2: Password update

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $password = $_POST['password'] ?? '';
    $confirmPassword = $_POST['confirmPassword'] ?? '';
    $strongPasswordRegex = "/^(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9])(?=.*[@!%*?&]).{6,}$/";

    if (empty($password) || empty($confirmPassword)) {
        $update_error = "Both fields are required.";
    } elseif ($password !== $confirmPassword) {
        $update_error = "Passwords do not match!";
    } elseif (!preg_match($strongPasswordRegex, $password)) {
        $update_error = "Password must be at least 6 characters long, include an uppercase letter, a lowercase letter, a number, and a special character!";
    } elseif (password_verify($password, $user['password_hash'])) {
        $update_error = "New password cannot be the same as your current password.";
    } else {
        $hashed_password = password_hash($password, PASSWORD_BCRYPT, ['cost' => 12]);
        $stmt = $pdo->prepare("UPDATE users SET password_hash = ?, reset_token = NULL, reset_expires_at = NULL WHERE email = ?");
        $stmt->execute([$hashed_password, $email]);

        unset($_SESSION['reset_email']);
        unset($_SESSION['reset_verified_email']);

        $update_success = "Password successfully reset! Redirecting to login...";
        header("Refresh: 2; url=login.php");
    }
}

include_once '../includes/header.php';
```

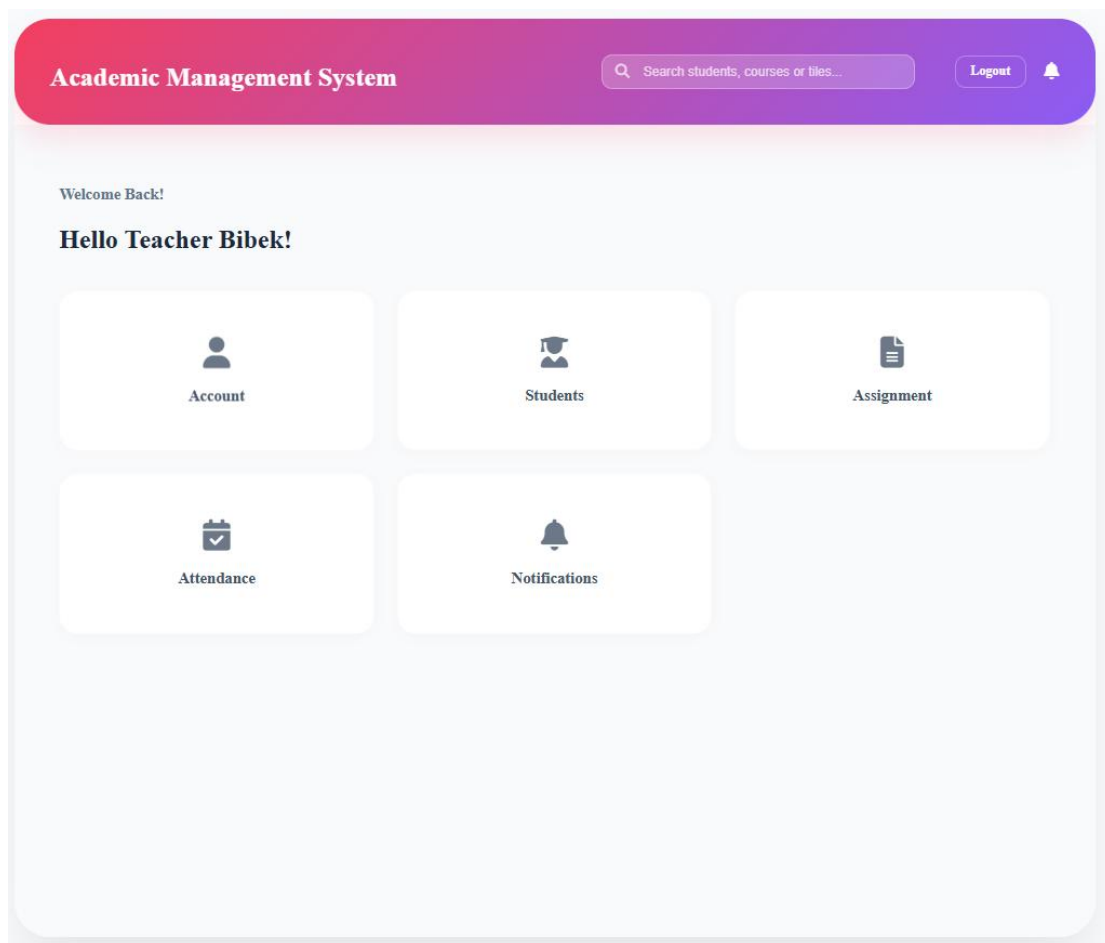
c) Teacher Dashboard & Student View (AMS-74 & AMS-81)

I was responsible for the frontend development of the portal used by the Teacher.

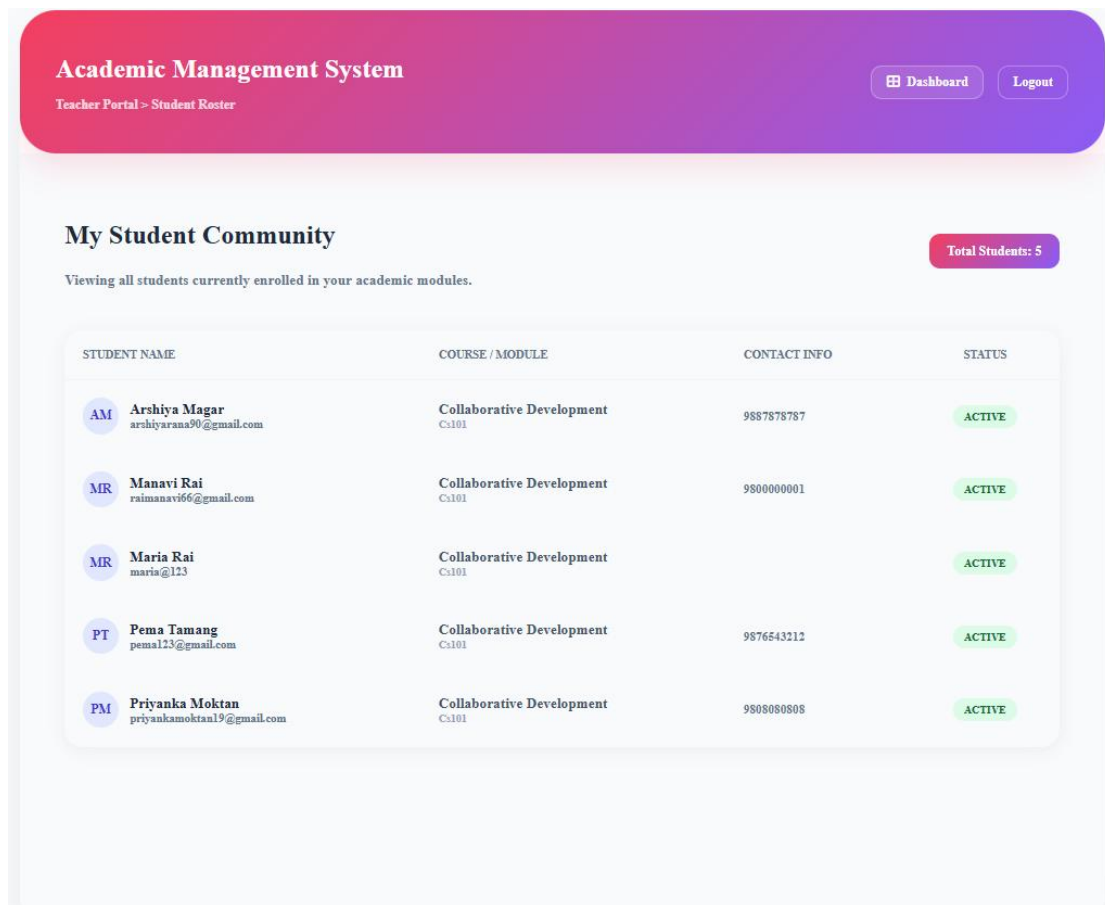
Dashboard Layout (AMS-74): I developed the tiled interface by implementing the CSS grid layout to make sure all tiles are perfectly aligned.

Student List View (AMS-81): I developed the students list page through looping with the help of the PHP foreach statement.

SCREENSHOT 3: Teacher Dashboard & Student Table



3.1:



CODE SCREENSHOT 3: SQL JOIN query

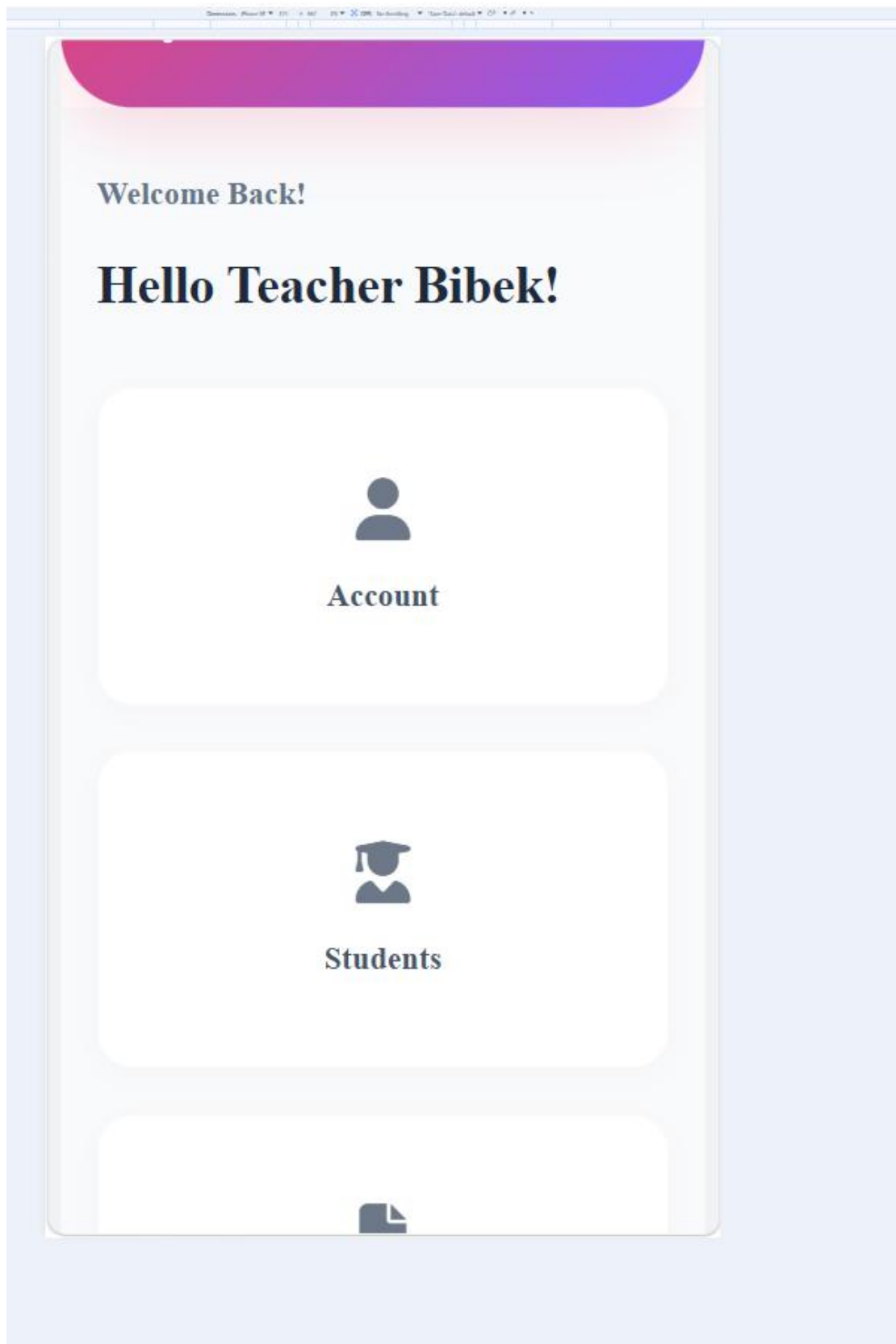
```
21  $stmt = $pdo->prepare("
22      SELECT DISTINCT u.first_name, u.last_name, u.email, u.contact, c.course_name, c.course_code
23      FROM enrollments e
24      JOIN students s ON e.student_id = s.id
25      JOIN users u ON s.user_id = u.id
26      JOIN courses c ON e.course_id = c.id
27      WHERE c.teacher_id = ?
28      ORDER BY c.course_name, u.first_name
29  ");
```

d) Responsive Design (AMS-76)

I made sure that the app would work properly on any device with CSS Media Queries (@media screen). My aim was to optimize the transparency on small devices through changes in the opacity of the background and vertical arrangement of navigation tiles.

I used `flex-direction: column` and `grid-template-columns: 1fr` to ensure the content "stacks" vertically on phones.

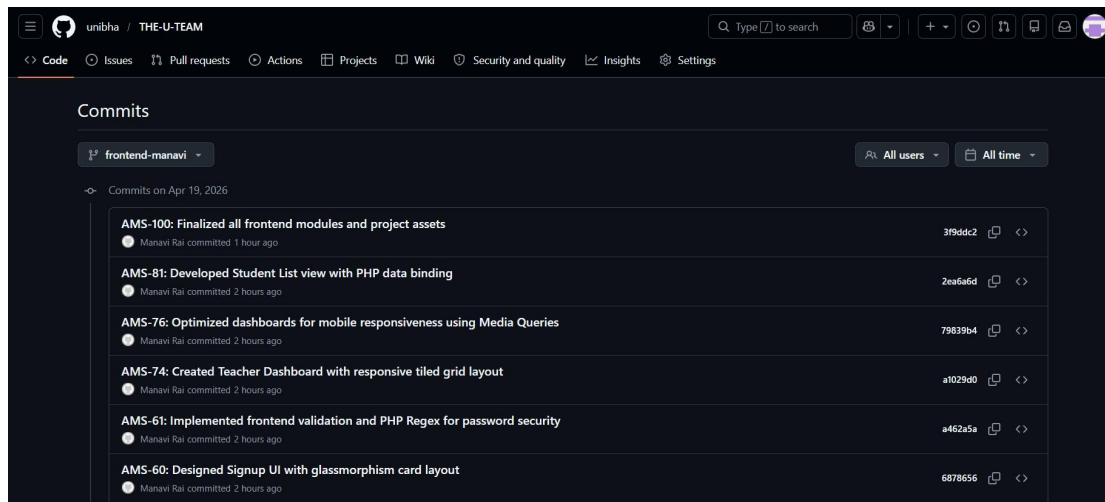
SCREENSHOT 4: Dashboard on Mobile



CODE SCREENSHOT 4: CSS Media Queries from style.css

```
@media (max-width: 768px) {  
  .form-body {  
    padding: 30px 20px;  
  }  
  
  .input-row {  
    flex-direction: column;  
    gap: 20px;  
  }  
  
  .role-selection {  
    flex-direction: column;  
  }  
  
  .auth-links-row {  
    flex-direction: column;  
    gap: 15px;  
  }  
  
  .login-link.divider {  
    border: none;  
    padding: 0;  
    border-top: 1px solid #e2e8f0;  
    padding-top: 15px;  
  }  
  
  .attendance-report-container {  
    grid-template-columns: 1fr;  
  }  
}  
  
@media (max-width: 640px) {  
  .dashboard-container {  
    flex-direction: column;  
  }  
  
  .sidebar {  
    width: 100%;  
    height: auto;  
  }  
}
```

e) Version Control Evidence:



The following screenshot shows the commit history of this project, indicating that I have used a modular and professional way of developing this project. The commits are mapped to individual issues in Jira by their unique ID (AMS-60 to AMS-81). This shows my continuous effort for the frontend development of this project during Sprint1.

5) Evidence of Collaboration & CDP

Jira Task Tracking:

Jira was used to monitor my work. This ensured clarity and made sure that I remained focused on what I had been tasked to do.

SCREENSHOT OF JIRA :

Spaces

THE U-TEAM

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

More 3

Search work

MR

AR

MM

PM

Filter 1

Group

Saved filters

...

☐	Work	Assignee	Reporter	Priority	Status	
☐	☒ AMS-81 View student list in teacher dashboard	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De
☐	☒ AMS-76 Make dashboards responsive	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De
☐	☒ AMS-74 Create Teacher Dashboard UI layout	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De
☐	☒ AMS-68 Create the new password input form UI	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De
☐	☒ AMS-67 Create the password recovery page UI	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De
☐	☒ AMS-61 Add frontend validation for signup fields	MR Manavi Rai	UN Unibha Neupane	High	DONE	De
☐	☒ AMS-60 Create signup page UI	MR Manavi Rai	UN Unibha Neupane	Medium	DONE	De

+ Create

7 of 7

Some issues occur during sprint 1:

is:issue state:closed

Labels

Milestones

New issue

☐ Open 0

☒ Closed 5

Author

Labels

Projects

Milestones

Assignees

Newest

☐	☒ Attendance page shows duplicate logs, incorrect percentage calculation, and inconsistent status entries
	#6 · by nijmi was closed 12 hours ago
☐	☒ UI Spacing Issues
	#5 · by nijmi was closed 12 hours ago
☐	☒ Reset Password Error
	#4 · by nijmi was closed 12 hours ago
☐	☒ OTP Validation Missing through mail
	#3 · by nijmi was closed 12 hours ago
☐	☒ Admin Login Failure
	#2 · by nijmi was closed 12 hours ago

[Give feedback](#)

Communication Evidence:

This Screenshot shows our team’s communication and how we coordinated the frontend and backend integration throughout the sprint1.

Thursday, March 26th ▾



Unibha Neupane 6:22 PM

@everyone coding kati vyayo?

march 29 ma cha time maile assign gareko

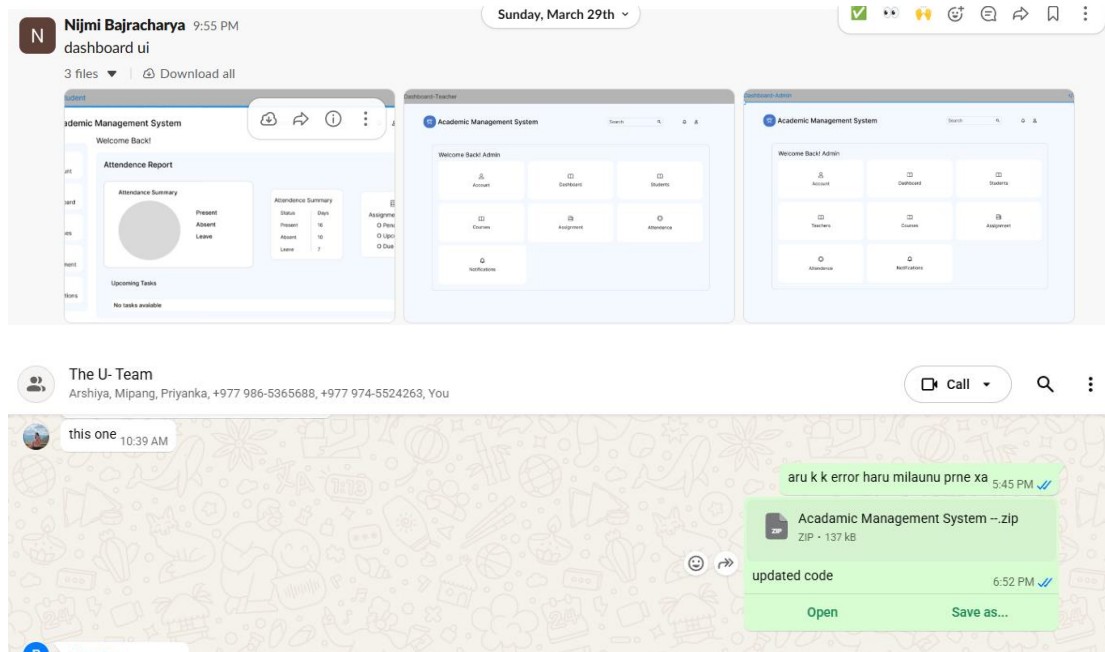
Friday, March 27th ▾



Rai Manavi 7:43 AM

complete bhaisky

push garisake github ma



Continuing Personal Development (CPD):

"In Sprint 1, I made great progress developing various competencies, both technically and professionally:

Technical Skills (CSS Grid & Flexbox): I knew about the existence of layouts before joining the sprint; however, working on the Teacher Dashboard (AMS-74) allowed me to learn to apply CSS Grid properly to achieve proper positioning of tiles. The

usage of grid-template-columns and flex-direction: column helped me to maintain the 'Glassmorphism' effect on cards of any screen resolution.

Professional Competence (Agile & Jira): One of the most significant experiences in the sprint was the transition to working within a team. I managed to understand how to break design requirements into tickets and commit them in Jira through unique Issue IDs, which gave me insight into traceability.

Advanced Debugging: Using the Browser Developer Tools is another valuable skill that has become my competence due to this sprint. With help from them, I was able to detect CSS issues and responsive 'break-points' to fix them in AMS-76 without modifying the core styling of the project.

Security-Focused Frontend Development: Implementing Regex validation (AMS-61) showed me that security is the first priority for any frontend developer since we deal with sensitive users' information."

6) References

GitHub: <https://github.com/unibha/THE-U-TEAM/tree/frontend-manavi>

Used for version control , code collaboration and tracking issues.

Jira: <https://heraldcollege-team-bkjr5ip.atlassian.net/jira/software/projects/AMS/list?>

For project management and tracking task

W3Schools : <https://www.w3schools.com>

For aesthetic references including Glassmorphism UI design techniques.

Google Fonts: <https://fonts.google.com>

For the Inter font family used across the AMS interface.

YouTube: <https://www.youtube.com>

For tutorial references on responsive design and CSS debugging techniques.